

Conducting Forensic Investigations on Linux Systems (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 06

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Time on Task:

2 hours, 41 minutes

Progress:

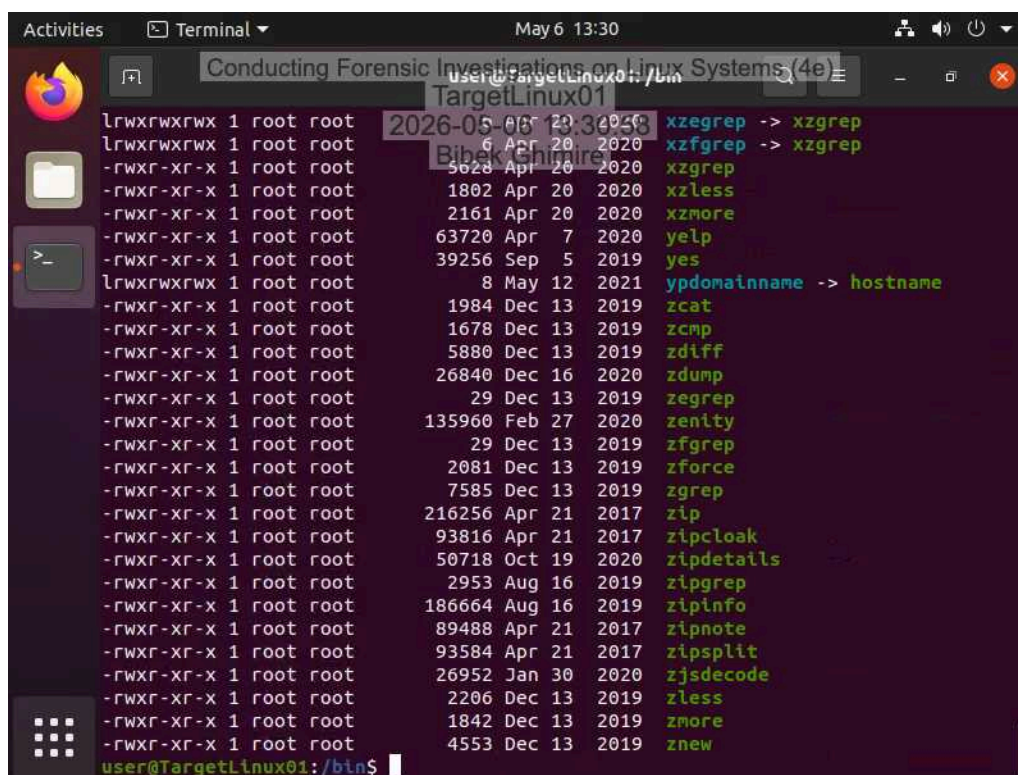
100%

Report Generated: Thursday, June 25, 2026 at 4:42 PM

Section 1: Hands-On Demonstration

Part 1: Explore a Live Linux System

17. Make a screen capture showing the contents of the `/bin` directory.



```
Activities Terminal May 6 13:30
Conducting Forensic Investigations on Linux Systems (4e)
user@TargetLinux01: /bin$ ls -la
lrwxrwxrwx 1 root root 20 Apr 20 2020 xzgrep -> xzgrep
lrwxrwxrwx 1 root root 20 Apr 20 2020 xzfgrep -> xzfgrep
-rwxr-xr-x 1 root root 5628 Apr 20 2020 xzgrep
-rwxr-xr-x 1 root root 1802 Apr 20 2020 xzless
-rwxr-xr-x 1 root root 2161 Apr 20 2020 xzmore
-rwxr-xr-x 1 root root 63720 Apr 7 2020 yelp
-rwxr-xr-x 1 root root 39256 Sep 5 2019 yes
lrwxrwxrwx 1 root root 8 May 12 2021 ypdomainname -> hostname
-rwxr-xr-x 1 root root 1984 Dec 13 2019 zcat
-rwxr-xr-x 1 root root 1678 Dec 13 2019 zcmp
-rwxr-xr-x 1 root root 5880 Dec 13 2019 zdiff
-rwxr-xr-x 1 root root 26840 Dec 16 2020 zdump
-rwxr-xr-x 1 root root 29 Dec 13 2019 zegrep
-rwxr-xr-x 1 root root 135960 Feb 27 2020 zenity
-rwxr-xr-x 1 root root 29 Dec 13 2019 zfgrep
-rwxr-xr-x 1 root root 2081 Dec 13 2019 zforce
-rwxr-xr-x 1 root root 7585 Dec 13 2019 zgrep
-rwxr-xr-x 1 root root 216256 Apr 21 2017 zip
-rwxr-xr-x 1 root root 93816 Apr 21 2017 zipcloak
-rwxr-xr-x 1 root root 50718 Oct 19 2020 zipdetails
-rwxr-xr-x 1 root root 2953 Aug 16 2019 zipgrep
-rwxr-xr-x 1 root root 186664 Aug 16 2019 zipinfo
-rwxr-xr-x 1 root root 89488 Apr 21 2017 zipnote
-rwxr-xr-x 1 root root 93584 Apr 21 2017 zipsplit
-rwxr-xr-x 1 root root 26952 Jan 30 2020 zjsdecode
-rwxr-xr-x 1 root root 2206 Dec 13 2019 zless
-rwxr-xr-x 1 root root 1842 Dec 13 2019 zmore
-rwxr-xr-x 1 root root 4553 Dec 13 2019 znew
user@TargetLinux01: /bin$
```

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 06

[illegible]

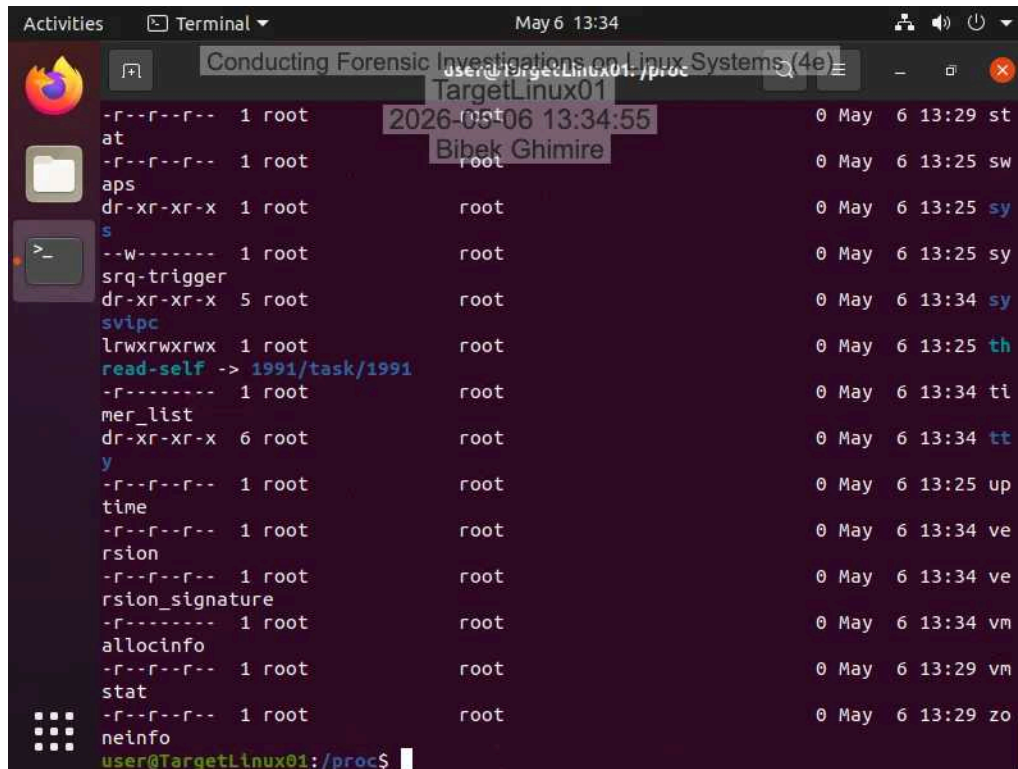
The screenshot shows a Kali Linux terminal window with the following content:

```

user@TargetLinux01: /var
ls -l
total 48
drwxr-xr-x 2 root root      4096 Jan  3  2022 backups
drwxr-xr-x 17 root root      4096 May 17  2021 cache
drwxrwsrwt 2 root whoopsie  4096 Apr 23  2020 crash
drwxr-xr-x 66 root root      4096 Aug 20  2021 lib
drwxrwsr-x 2 root staff     4096 Apr 15  2020 local
lrwxrwxrwx 1 root root         9 May 12  2021 lock -> /run/lock
drwxrwxr-x 13 root syslog   4096 May  6 13:25 log
drwxrwsr-x 2 root mail      4096 Apr 23  2020 mail
drwxrwsrwt 2 root whoopsie  4096 Apr 23  2020 metrics
drwxr-xr-x 2 root root      4096 Apr 23  2020 opt
lrwxrwxrwx 1 root root         4 May 12  2021 run -> /run
drwxr-xr-x 12 root root      4096 Jan  3  2022 snap
drwxr-xr-x 7 root root       4096 Aug 20  2021 spool
drwxrwsrwt 10 root root      4096 May  6 13:31 tmp

```

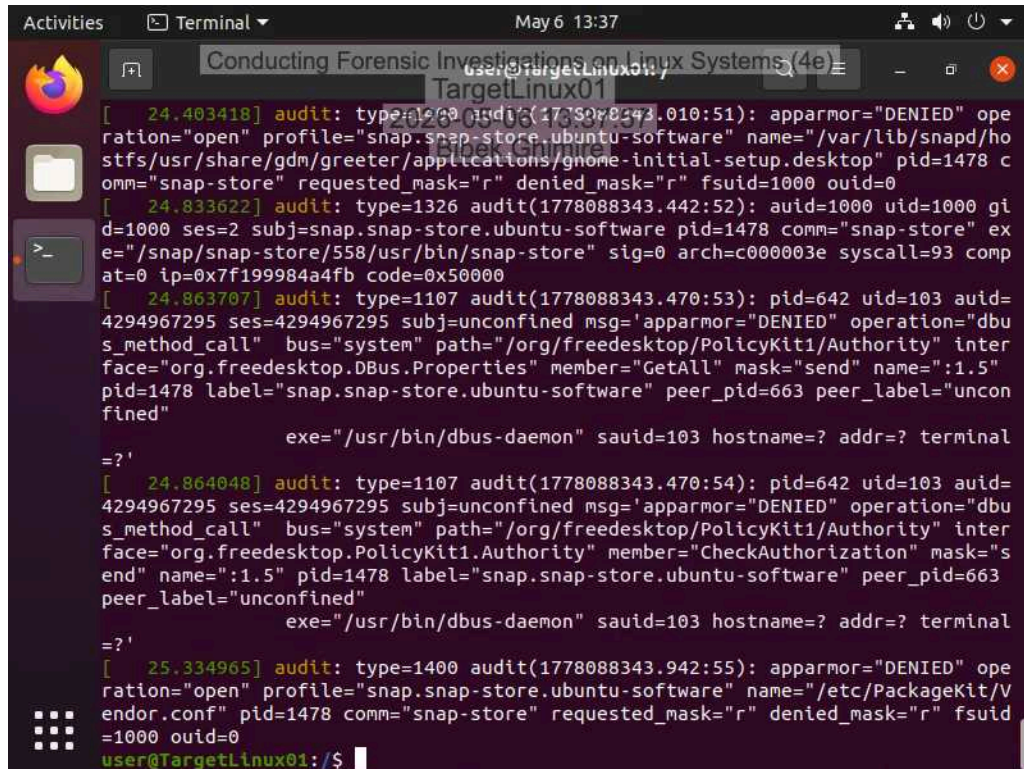
22. Make a screen capture showing the contents of the /proc directory.



```
user@TargetLinux01: /proc$ ls -la
-r--r--r-- 1 root      0 May  6 13:29 stat
-r--r--r-- 1 root      0 May  6 13:25 sw
-r--r--r-- 1 root      0 May  6 13:25 sy
dr-xr-xr-x 1 root      0 May  6 13:25 sy
--w----- 1 root      0 May  6 13:25 sy
srq-trigger
dr-xr-xr-x 5 root      0 May  6 13:34 sy
svipc
lrwxrwxrwx 1 root      0 May  6 13:25 th
read-self -> 1991/task/1991
-r----- 1 root      0 May  6 13:34 ti
mer_list
dr-xr-xr-x 6 root      0 May  6 13:34 tt
y
-r--r--r-- 1 root      0 May  6 13:25 up
time
-r--r--r-- 1 root      0 May  6 13:34 ve
rsion
-r--r--r-- 1 root      0 May  6 13:34 ve
rsion_signature
-r----- 1 root      0 May  6 13:34 vm
allocinfo
-r--r--r-- 1 root      0 May  6 13:29 vm
stat
-r--r--r-- 1 root      0 May  6 13:29 zo
neinfo
```

Part 2: Use Linux Shell Commands for Forensic Investigations

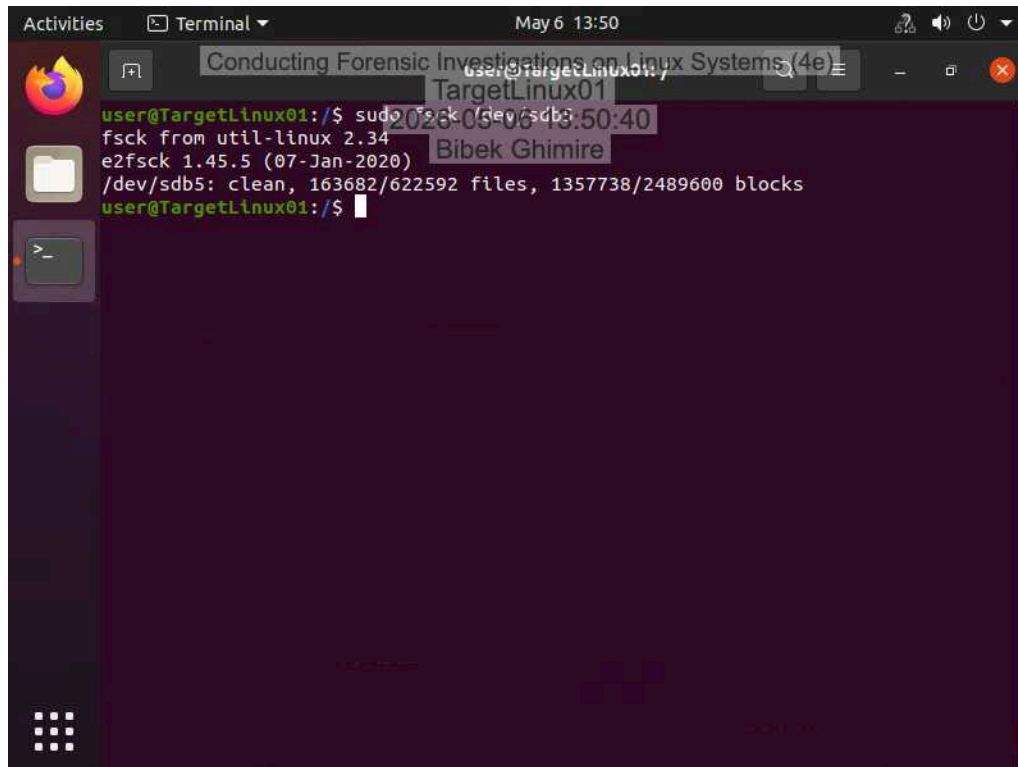
2. Make a screen capture showing the results of the `dmesg` command.



The screenshot shows a terminal window titled "Terminal" with the date and time "May 6 13:37". The terminal displays the output of the `dmesg` command, showing several audit messages. The messages are formatted as follows:

```
[ 24.403418] audit: type=1600 audit(1778088343.010:51): apparmor="DENIED" operation="open" profile="snap.snap-store.ubuntu-software" name="/var/lib/snapd/hosts/usr/share/gdm/greeter/applications/gnome-initial-setup.desktop" pid=1478 comm="snap-store" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[ 24.833622] audit: type=1326 audit(1778088343.442:52): auid=1000 uid=1000 gid=1000 ses=2 subj=snap.snap-store.ubuntu-software pid=1478 comm="snap-store" exe="/snap/snap-store/558/usr/bin/snap-store" sig=0 arch=c000003e syscall=93 compat=0 ip=0x7f199984a4fb code=0x50000
[ 24.863707] audit: type=1107 audit(1778088343.470:53): pid=642 uid=103 auid=4294967295 ses=4294967295 subj=unconfined msg='apparmor="DENIED" operation="dbus_method_call" bus="system" path="/org/freedesktop/PolicyKit1/Authority" interface="org.freedesktop.DBus.Properties" member="GetAll" mask="send" name=":1.5" pid=1478 label="snap.snap-store.ubuntu-software" peer_pid=663 peer_label="unconfined" exe="/usr/bin/dbus-daemon" sauid=103 hostname=? addr=? terminal=?'
[ 24.864048] audit: type=1107 audit(1778088343.470:54): pid=642 uid=103 auid=4294967295 ses=4294967295 subj=unconfined msg='apparmor="DENIED" operation="dbus_method_call" bus="system" path="/org/freedesktop/PolicyKit1/Authority" interface="org.freedesktop.PolicyKit1.Authority" member="CheckAuthorization" mask="send" name=":1.5" pid=1478 label="snap.snap-store.ubuntu-software" peer_pid=663 peer_label="unconfined" exe="/usr/bin/dbus-daemon" sauid=103 hostname=? addr=? terminal=?'
[ 25.334965] audit: type=1400 audit(1778088343.942:55): apparmor="DENIED" operation="open" profile="snap.snap-store.ubuntu-software" name="/etc/PackageKit/Vendor.conf" pid=1478 comm="snap-store" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
user@TargetLinux01:/$
```

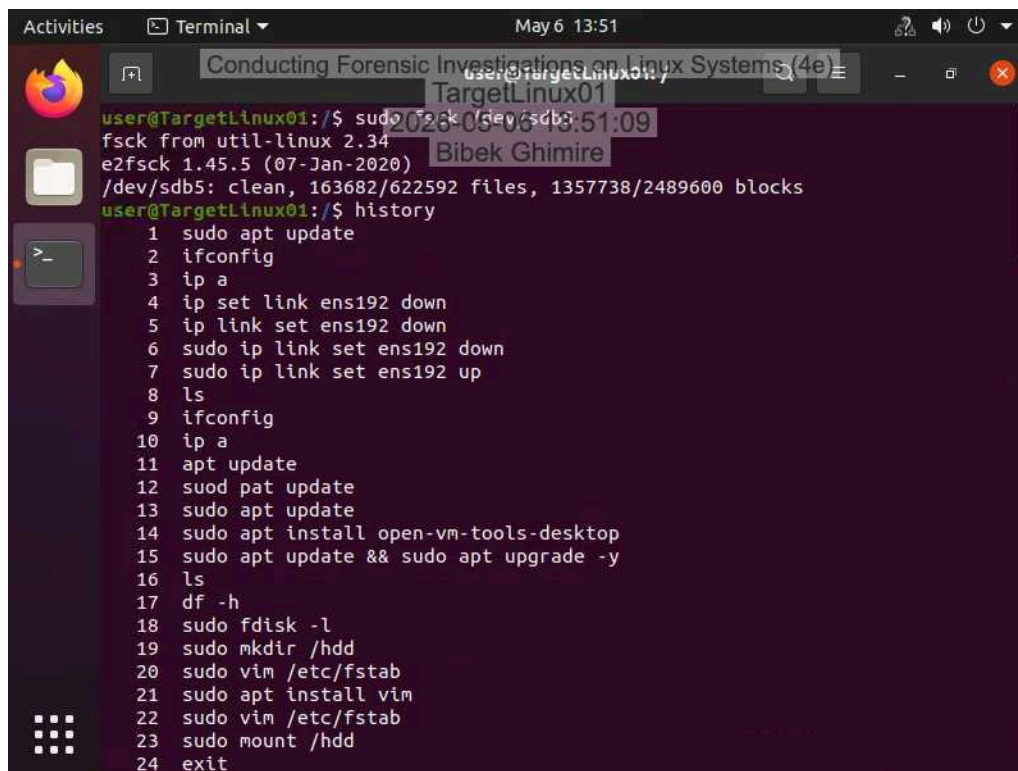
7. Make a screen capture showing the results of the fsck command.



A terminal window titled "Conducting Forensic Investigations on Linux Systems (4e)" showing the execution of the fsck command. The prompt is "user@TargetLinux01:/\$". The command "sudo fsck /dev/sdb5" is entered. The output shows the fsck process running on /dev/sdb5, reporting it as clean with 163682/622592 files and 1357738/2489600 blocks. The prompt returns to "user@TargetLinux01:/\$".

```
user@TargetLinux01:/$ sudo fsck /dev/sdb5
fsck from util-linux 2.34
e2fsck 1.45.5 (07-Jan-2020)
/dev/sdb5: clean, 163682/622592 files, 1357738/2489600 blocks
user@TargetLinux01:/$
```

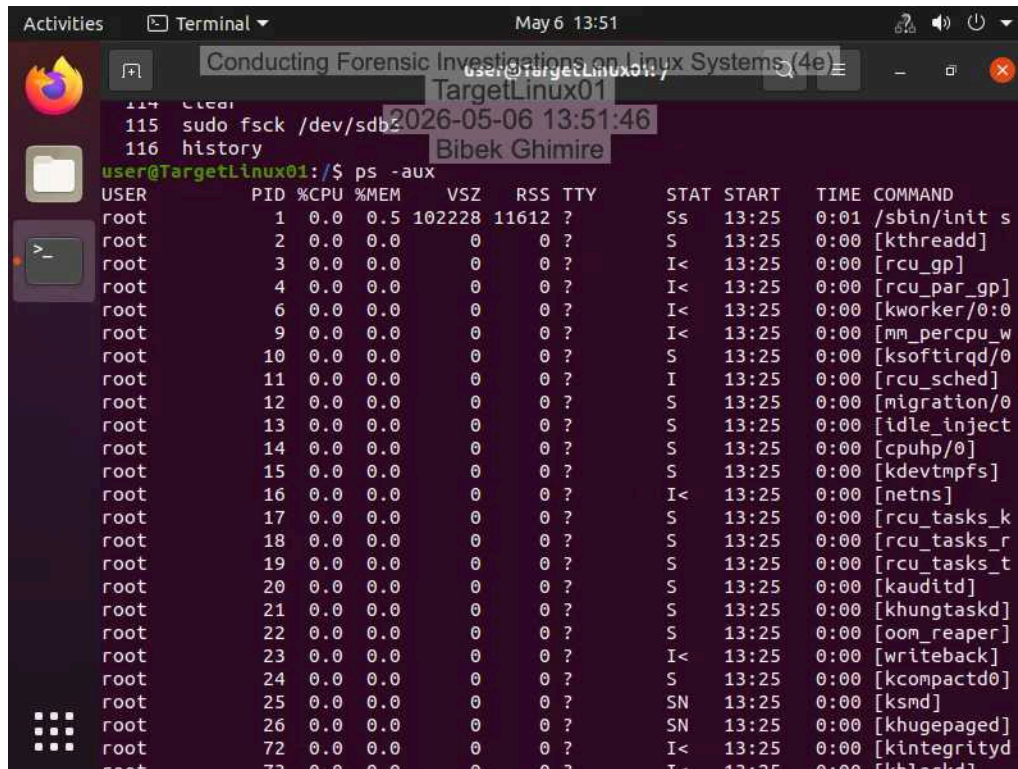
9. Make a screen capture showing the results of the history command.



A terminal window titled "Conducting Forensic Investigations on Linux Systems (4e)" showing the results of the history command. The prompt is "user@TargetLinux01:/\$". The command "history" is entered. The output lists 24 commands executed in the session, including updates, network configuration, disk management, and file installation. The prompt returns to "user@TargetLinux01:/\$".

```
user@TargetLinux01:/$ history
1  sudo apt update
2  ifconfig
3  ip a
4  ip set link ens192 down
5  ip link set ens192 down
6  sudo ip link set ens192 down
7  sudo ip link set ens192 up
8  ls
9  ifconfig
10 ip a
11 apt update
12 suod pat update
13 sudo apt update
14 sudo apt install open-vm-tools-desktop
15 sudo apt update && sudo apt upgrade -y
16 ls
17 df -h
18 sudo fdisk -l
19 sudo mkdir /hdd
20 sudo vim /etc/fstab
21 sudo apt install vim
22 sudo vim /etc/fstab
23 sudo mount /hdd
24 exit
```

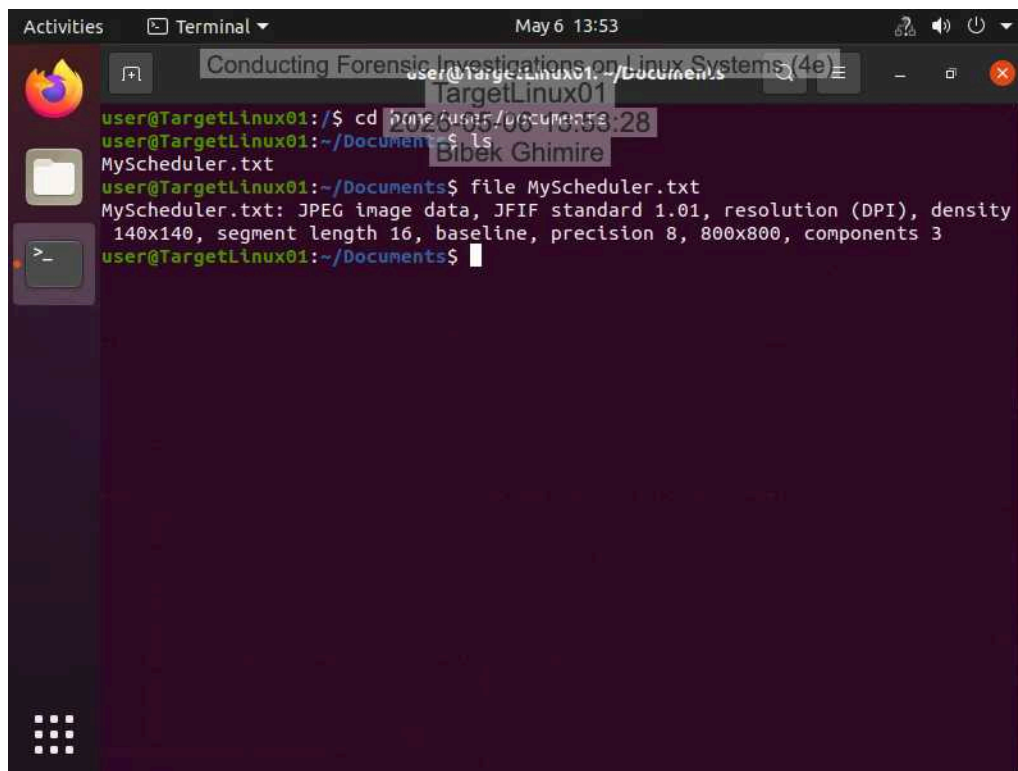
11. Make a screen capture showing the running processes.



A terminal window on a Linux system showing the output of the `ps -aux` command. The window title is "Conducting Forensic Investigations on Linux Systems (4e)". The terminal shows the user `user@TargetLinux01` and the time `2026-05-06 13:51:46`. The output lists various system processes including `/sbin/init`, `[kthreadd]`, `[rcu_gp]`, `[rcu_par_gp]`, `[kworker/0:0]`, `[mm_percpu_w]`, `[ksoftirqd/0]`, `[migration/0]`, `[idle_inject]`, `[cpuhp/0]`, `[kdevtmpfs]`, `[netns]`, `[rcu_tasks_k]`, `[rcu_tasks_r]`, `[rcu_tasks_t]`, `[kauditd]`, `[khungtaskd]`, `[oom_reaper]`, `[writeback]`, `[kcompactd0]`, `[ksmd]`, `[khugepaged]`, `[kintegrityd]`, and `[kblockd]`.

```
user@TargetLinux01:/$ ps -aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.5 102228 11612 ?        Ss   13:25   0:01 /sbin/init s
root           2  0.0  0.0      0     0 ?        S    13:25   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        I<   13:25   0:00 [rcu_gp]
root           4  0.0  0.0      0     0 ?        I<   13:25   0:00 [rcu_par_gp]
root           6  0.0  0.0      0     0 ?        I<   13:25   0:00 [kworker/0:0]
root           9  0.0  0.0      0     0 ?        I<   13:25   0:00 [mm_percpu_w]
root          10  0.0  0.0      0     0 ?        S    13:25   0:00 [ksoftirqd/0]
root          11  0.0  0.0      0     0 ?        I    13:25   0:00 [rcu_sched]
root          12  0.0  0.0      0     0 ?        S    13:25   0:00 [migration/0]
root          13  0.0  0.0      0     0 ?        S    13:25   0:00 [idle_inject]
root          14  0.0  0.0      0     0 ?        S    13:25   0:00 [cpuhp/0]
root          15  0.0  0.0      0     0 ?        S    13:25   0:00 [kdevtmpfs]
root          16  0.0  0.0      0     0 ?        I<   13:25   0:00 [netns]
root          17  0.0  0.0      0     0 ?        S    13:25   0:00 [rcu_tasks_k]
root          18  0.0  0.0      0     0 ?        S    13:25   0:00 [rcu_tasks_r]
root          19  0.0  0.0      0     0 ?        S    13:25   0:00 [rcu_tasks_t]
root          20  0.0  0.0      0     0 ?        S    13:25   0:00 [kauditd]
root          21  0.0  0.0      0     0 ?        S    13:25   0:00 [khungtaskd]
root          22  0.0  0.0      0     0 ?        S    13:25   0:00 [oom_reaper]
root          23  0.0  0.0      0     0 ?        I<   13:25   0:00 [writeback]
root          24  0.0  0.0      0     0 ?        S    13:25   0:00 [kcompactd0]
root          25  0.0  0.0      0     0 ?        SN   13:25   0:00 [ksmd]
root          26  0.0  0.0      0     0 ?        SN   13:25   0:00 [khugepaged]
root          72  0.0  0.0      0     0 ?        I<   13:25   0:00 [kintegrityd]
root          73  0.0  0.0      0     0 ?        I<   13:25   0:00 [kblockd]
```

15. Make a screen capture showing the results of the file command.

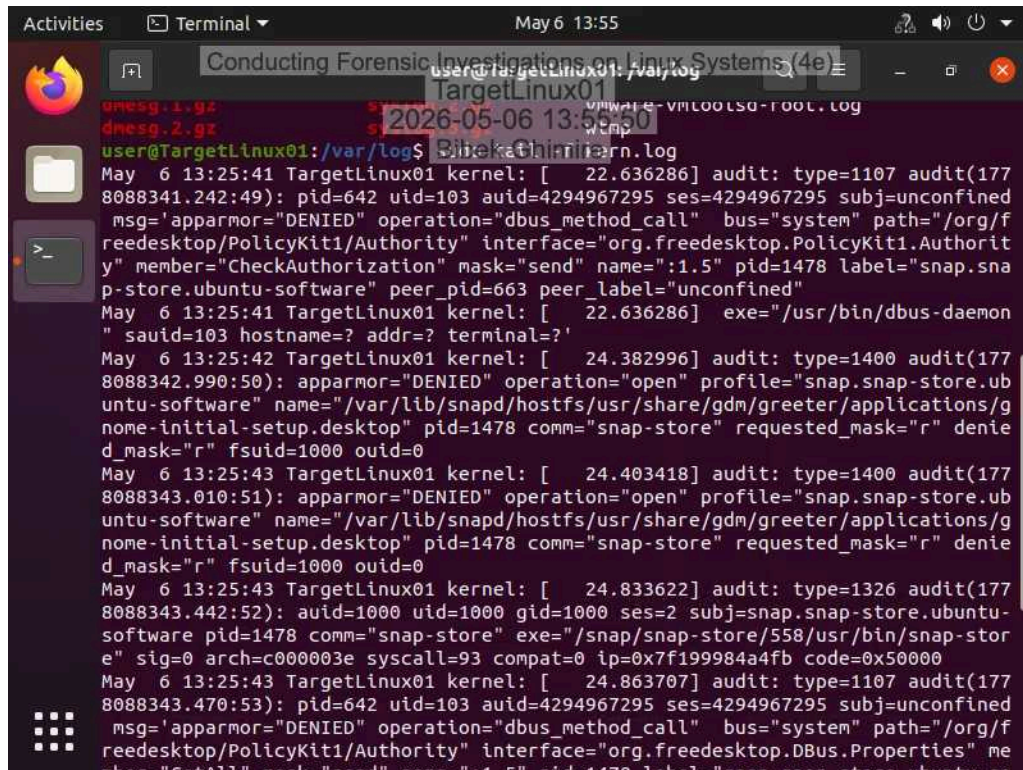


A terminal window on a Linux system showing the output of the `file` command. The window title is "Conducting Forensic Investigations on Linux Systems (4e)". The terminal shows the user `user@TargetLinux01` and the time `2026-05-06 13:53:28`. The user navigates to the `~/Documents` directory and runs `file MyScheduler.txt`. The output indicates that `MyScheduler.txt` is a JPEG image data, JFIF standard 1.01, resolution (DPI), density 140x140, segment length 16, baseline, precision 8, 800x800, components 3.

```
user@TargetLinux01:/$ cd ~/Documents
user@TargetLinux01:~/Documents$ ls
MyScheduler.txt
user@TargetLinux01:~/Documents$ file MyScheduler.txt
MyScheduler.txt: JPEG image data, JFIF standard 1.01, resolution (DPI), density
140x140, segment length 16, baseline, precision 8, 800x800, components 3
user@TargetLinux01:~/Documents$
```

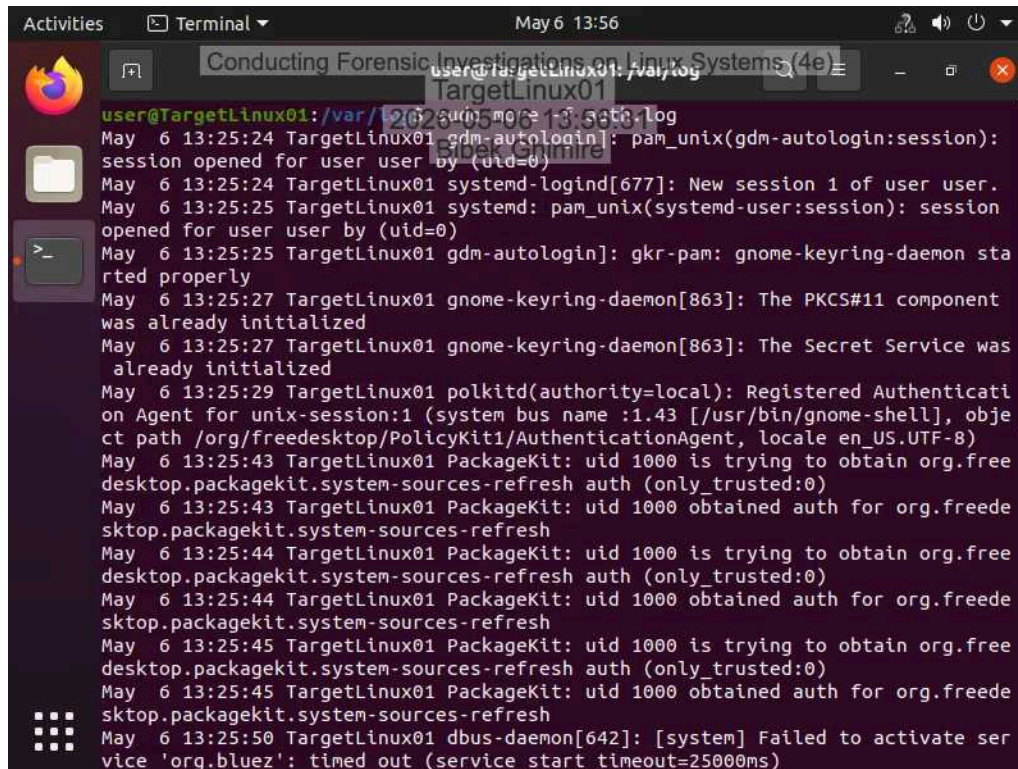

Part 3: Retrieve Logs Files on a Live Linux System

4. Make a screen capture showing the records in the kern.log file.



```
Activities Terminal May 6 13:55
Conducting Forensic Investigations on Linux Systems (4e)
TargetLinux01
user@TargetLinux01: /var/log$ cat /var/log/kern.log
May 6 13:25:41 TargetLinux01 kernel: [ 22.636286] audit: type=1107 audit(1778088341.242:49): pid=642 uid=103 auid=4294967295 ses=4294967295 subj=unconfined msg='apparmor="DENIED" operation="dbus_method_call" bus="system" path="/org/freedesktop/PolicyKit1/Authority" interface="org.freedesktop.PolicyKit1.Authority" member="CheckAuthorization" mask="send" name=":1.5" pid=1478 label="snap.snap-store.ubuntu-software" peer_pid=663 peer_label="unconfined"'
May 6 13:25:41 TargetLinux01 kernel: [ 22.636286] exe="/usr/bin/dbus-daemon" sauid=103 hostname=? addr=? terminal=?
May 6 13:25:42 TargetLinux01 kernel: [ 24.382996] audit: type=1400 audit(1778088342.990:50): apparmor="DENIED" operation="open" profile="snap.snap-store.ubuntu-software" name="/var/lib/snapd/hostfs/usr/share/gdm/greeter/applications/gnome-initial-setup.desktop" pid=1478 comm="snap-store" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
May 6 13:25:43 TargetLinux01 kernel: [ 24.403418] audit: type=1400 audit(1778088343.010:51): apparmor="DENIED" operation="open" profile="snap.snap-store.ubuntu-software" name="/var/lib/snapd/hostfs/usr/share/gdm/greeter/applications/gnome-initial-setup.desktop" pid=1478 comm="snap-store" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
May 6 13:25:43 TargetLinux01 kernel: [ 24.833622] audit: type=1326 audit(1778088343.442:52): auid=1000 uid=1000 gid=1000 ses=2 subj=snap.snap-store.ubuntu-software pid=1478 comm="snap-store" exe="/snap/snap-store/558/usr/bin/snap-store" sig=0 arch=c000003e syscall=93 compat=0 ip=0x7f199984a4fb code=0x50000
May 6 13:25:43 TargetLinux01 kernel: [ 24.863707] audit: type=1107 audit(1778088343.470:53): pid=642 uid=103 auid=4294967295 ses=4294967295 subj=unconfined msg='apparmor="DENIED" operation="dbus_method_call" bus="system" path="/org/freedesktop/PolicyKit1/Authority" interface="org.freedesktop.DBus.Properties" member="GetAll" mask="send" name=":1.5" pid=1478 label="snap.snap-store.ubuntu-software" peer_pid=663 peer_label="unconfined"'
```

7. Make a screen capture showing the records in the auth.log file.

A terminal window titled 'Conducting Forensic Investigations on Linux Systems (4e)' is shown. The terminal displays the contents of the /var/log/auth.log file. The output shows a series of system logs from May 6, 2025, at 13:25. The logs include session opening for user 'user', systemd-logind session creation, gdm-autologin success, gnome-keyring-daemon startup, polkitd authentication, and PackageKit system-source refresh attempts. The terminal window has a dark background with light-colored text. The top of the window shows the title bar with 'Activities', 'Terminal', and the time 'May 6 13:56'. The left sidebar shows icons for the terminal, a file manager, and a web browser. The right sidebar shows icons for a search, a window manager, and a power button.

```
user@TargetLinux01: /var/log
user@TargetLinux01: ~$ cat /var/log/auth.log
May  6 13:25:24 TargetLinux01 gdm-autologin[13536]: pam_unix(gdm-autologin:session): session opened for user user by (uid=0)
May  6 13:25:24 TargetLinux01 systemd-logind[677]: New session 1 of user user.
May  6 13:25:25 TargetLinux01 systemd: pam_unix(systemd-user:session): session opened for user user by (uid=0)
May  6 13:25:25 TargetLinux01 gdm-autologin]: gkr-pam: gnome-keyring-daemon started properly
May  6 13:25:27 TargetLinux01 gnome-keyring-daemon[863]: The PKCS#11 component was already initialized
May  6 13:25:27 TargetLinux01 gnome-keyring-daemon[863]: The Secret Service was already initialized
May  6 13:25:29 TargetLinux01 polkitd(authority=local): Registered Authentication Agent for unix-session:1 (system bus name :1.43 [/usr/bin/gnome-shell], object path /org/freedesktop/PolicyKit1/AuthenticationAgent, locale en_US.UTF-8)
May  6 13:25:43 TargetLinux01 PackageKit: uid 1000 is trying to obtain org.freedesktop.packagekit.system-sources-refresh auth (only_trusted:0)
May  6 13:25:43 TargetLinux01 PackageKit: uid 1000 obtained auth for org.freedesktop.packagekit.system-sources-refresh
May  6 13:25:44 TargetLinux01 PackageKit: uid 1000 is trying to obtain org.freedesktop.packagekit.system-sources-refresh auth (only_trusted:0)
May  6 13:25:44 TargetLinux01 PackageKit: uid 1000 obtained auth for org.freedesktop.packagekit.system-sources-refresh
May  6 13:25:45 TargetLinux01 PackageKit: uid 1000 is trying to obtain org.freedesktop.packagekit.system-sources-refresh auth (only_trusted:0)
May  6 13:25:45 TargetLinux01 PackageKit: uid 1000 obtained auth for org.freedesktop.packagekit.system-sources-refresh
May  6 13:25:50 TargetLinux01 dbus-daemon[642]: [system] Failed to activate service 'org.bluez': timed out (service_start_timeout=25000ms)
```


Section 2: Applied Learning

Part 1: Identify Login Attempts on a Linux Drive Image

15. **Document** the names of the two non-root users that attempted to log in, the number of attempts detected, the date/time range of the attempts, the source IP address for the login attempts, and the port.

Dominic and Noel were non-root users that attempted to log in 22 times from source ip: 192.168.78.1 port 14444 and port 3417 on the June 11 from the hours of 00:57:08 to 05:39:01

17. **Document** the date and time the most recent successful login for the user(s) that you previously identified in step 15.

Latest successful login from "dominic" was Jun 11 05:23:03 and for other user "noel", he was unable to successfully get into the system.

Part 2: Identify Software Installations on a Linux Drive Image

3. **Document** the applications that were installed using apt-get, then use the Internet to identify the ones that might be considered suspicious.

Logkeys, autotools-dev, build-essential, autoconf, kbd, cacti, and openssh-server are the applications installed using apt-get. Logkeys, in particular, can be used to gain remote access to a system and serve as a backdoor. On the other hand, cacti is a network monitoring tool that has vulnerabilities that allow remote code execution (RCE).

Part 3: Identify External Drive Attachments on a Linux Drive Image

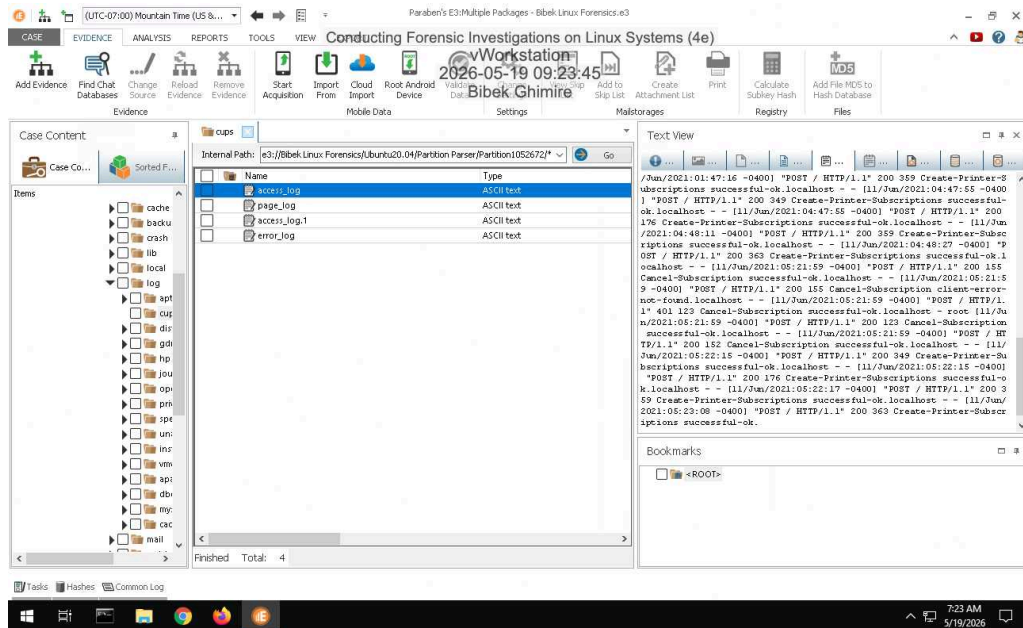
4. **Document** when the USB storage device was connected and its serial number.

USB storage device was connected on Jun 10 10:24:12 and serial number : FBI1405291710344

Section 3: Challenge and Analysis

Part 1: Identify Recently Printed Files on a Linux Drive Image

Make a screen capture showing the contents of the printer log file.



Part 2: Identify Disk Imaging on a Linux Drive Image

Make a screen capture showing the record of the dd command in the Text View.

